Toward a Constructive Theory of Radicals

A Critical and Synthetic Inquiry of
Ontological Foundations, Ghosh's Quintic Parametrisation, and Interdisciplinary Perspectives

Soumadeep Ghosh and Claude (Anthropic)

Kolkata, India

Abstract

We argue that the classical impossibility theorems of Abel–Ruffini and Galois theory, while mathematically correct within their own frameworks, rest on an ontological foundation that has never been fully examined: the definition of the *power function* and hence of *radical expressions*. We show, through the example $(-\frac{1}{2})^{-\frac{1}{2}}$, that every standard definition of a radical silently presupposes a branch choice, rendering "solvability by radicals" an inherently ambiguous predicate unless branch choices are made explicit and constructive. We then construct, from first principles and without borrowing ontological furniture from Galois theory, a *Constructive Theory of Radicals* ($\mathcal{R}$-theory) in which radical expressions are defined as explicitly branched, finite-depth objects over a ground ring. Within this framework we re-examine Ghosh's quintic parametrisation [7, 8, 9, 10]: an algorithm (Algorithm G) that decomposes any monic quintic into a quadratic factor and a cubic factor, each solvable by radicals. We prove that Galois's theorem, read as an impossibility result about *uniform symbolic formulas*, does not preclude Ghosh's *constructive method*, and we invoke Gauss's theorem on polynomial factorisation to show that the residual cubic after extracting the quadratic roots has coefficients in the original coefficient field, eliminating the need for complex extensions. We further observe that Cardano's formula is not generically required: when the residual cubic has one real root and two complex conjugate roots—the generic case—the real root may be extracted constructively, reducing the cubic to a quadratic solvable by radicals. We situate these findings in economics, finance, game theory, philosophy, political science, geopolitics, warfare economics, statistics, computer science, and artificial intelligence.

The paper ends with "The End"

Contents

1 Introduction

The question of solving polynomial equations in radicals is among the oldest and most philosophically charged in mathematics. The quadratic formula is classical. Cardano published solutions to the cubic and quartic in the sixteenth century, though the cubic method originated with Scipione del Ferro and was independently rediscovered by Tartaglia—a historical fact that carries, as we shall argue, profound ontological weight. Abel [1] and Galois [5] then established that no *uniform* radical formula exists for the general quintic, a result that has dominated the landscape for nearly two centuries.

Ghosh's series of papers [7, 8, 9, 10] offers a constructive counterpoint. Rather than seeking a single formula valid simultaneously for all quintics, Ghosh demonstrates that for any *given* quintic one may choose auxiliary parameters P, Q such that the polynomial factors into a cubic times a quadratic, each factor then solvable in radicals. The approach is presented as consistent with Abel–Ruffini.

Our contribution is threefold.

(i) We identify a foundational gap in the classical theory: the definition of "solvability in radicals" depends on the *power function*, which is not well-defined as a primitive operation. The expression $(-\frac{1}{2})^{-\frac{1}{2}}$ makes this concrete: it admits multiple values depending on branch choices that the classical theory leaves implicit.

(ii) We construct, from first principles, a *Constructive Theory of Radicals* that makes branch choices explicit and defines radical expressions as finitely generated, explicitly branched objects. Within this theory Galois's impossibility result is reinterpreted as applying to *formulas*, not to *methods*.

(iii) We argue that Ghosh's Algorithm G, evaluated on its own ontological terms, is a method in the sense of del Ferro and Tartaglia—not a formula in the sense of Cardano—and that the constructive theory we develop provides the appropriate framework for its evaluation.

Organisation. Section 2 develops the ontological critique of the power function. Section 3 constructs the Constructive Theory of Radicals. Section 4 distinguishes formulas from methods. Section 5 reviews Ghosh's algebraic machinery. Section 6 proves the existence theorem. Section 7 applies Gauss's theorem to eliminate complex extensions. Section 8 analyses when Cardano is and is not needed. Section 9 examines logical and proof-theoretic aspects. Sections 10–15 develop interdisciplinary perspectives. Section 16 presents Algorithm G with complexity analysis. Section 18 gives the geometric interpretation. Section 19 concludes.

2 The Ontological Crisis of the Power Function

2.1 The defining expression

Consider the expression

$$(-\frac{1}{2})^{-\frac{1}{2}}.$$

Unfolding the definition:

$$(-\frac{1}{2})^{-1/2} = \frac{1}{(-\frac{1}{2})^{1/2}} = \frac{1}{\sqrt{-\frac{1}{2}}}.$$

Now $\sqrt{-\frac{1}{2}} = \sqrt{-1} \cdot \sqrt{\frac{1}{2}}$, which requires choosing a square root of -1 (either i or $-i$) and a square root of $\frac{1}{2}$ (either $\frac{1}{\sqrt{2}}$ or $-\frac{1}{\sqrt{2}}$). The four combinations yield four values:

$$(-\frac{1}{2})^{-1/2} \in \{-i\sqrt{2},\ i\sqrt{2},\ i\sqrt{2},\ -i\sqrt{2}\} = \{\pm i\sqrt{2}\}.$$

The expression has *two values*, not one. The classical convention selects the *principal branch* — but that convention is not part of the expression itself. It is an external, ontologically invisible choice.

2.2 The general problem

Definition 2.1 (Power function, classical). *For $x \in \mathbb{C} \setminus \{0\}$ and $y \in \mathbb{C}$, the power function is defined as*

$$x^y := e^{y \log x},$$

where $\log x = \ln|x| + i\ \arg(x)$ *with* $\arg(x) \in (-\pi, \pi]$ *(principal branch).*

This definition contains a suppressed choice: $\arg(x)$ is selected from the set $\{\theta + 2\pi k : k \in \mathbb{Z}\}$. Different branches yield different values of x^y. Specifically, $x^{p/q}$ has q distinct values, and classical radical expressions implicitly select one.

Proposition 2.2. *Every classical radical expression α of depth n over a field F implicitly encodes a sequence of n branch choices, none of which is determined by α alone.*

Proof. By induction on depth. At depth 1, $\alpha = a^{1/k}$ for $a \in F$, $k \in \mathbb{N}$, which has k values in $\mathbb{C}$; the choice among them is not encoded in the expression. At depth n, $\alpha = \beta^{1/k}$ where β is a depth-$(n-1)$ expression; by induction β depends on $n-1$ branch choices, and α adds one more. □

2.3 Galois theory's response and its limits

Galois theory handles multivaluedness by *abstracting over* all branch choices simultaneously: the Galois group $\mathrm{Gal}(K/F)$ is precisely the group of automorphisms encoding all consistent reassignments of branch choices. This is mathematically powerful but ontologically evasive: it does not *resolve* the ambiguity, it *packages* it.

Remark 2.3. *The impossibility theorem of Abel–Ruffini, as formalised in Galois theory, is a theorem about the non-existence of a* formula *— a uniform symbolic expression invariant under all branch choices simultaneously (i.e., lying in a radical tower over $\mathbb{Q}(a, b, c, d, e)$). It is not, strictly speaking, a theorem about the non-existence of* constructive procedures *that make specific branch choices explicitly.*

3 A Constructive Theory of Radicals

We now construct a Theory of Radicals ($\mathcal{R}$-theory) from first principles, without reference to field extensions, Galois groups, or the classical power function.

3.1 Primitive objects

Axiom 3.1 (Ground ring)**.** *Fix a computable commutative ring F_0 (e.g., $\mathbb{Q}$ or $\mathbb{Z}$). Elements of F_0 are the* ground objects*.*

Axiom 3.2 (Primitive radical operation)**.** *A* primitive radical operation *of index $n \in \mathbb{N}$, $n \geq 2$, applied to $a \in F_0$, produces a symbol $\langle a, n, j\rangle$ for $j \in \{0, 1, \ldots, n-1\}$, representing the j-th root of $x^n - a = 0$ in some fixed ordering. The* branch index *j is an explicit part of the symbol.*

3.2 The radical hierarchy

Definition 3.3 (Radical expressions, $\mathcal{R}_n$)**.** *Define $\mathcal{R}_0 := F_0$. For $n \geq 1$, define $\mathcal{R}_n$ as the closure of $\mathcal{R}_{n-1} \cup \{\langle a, k, j\rangle : a \in \mathcal{R}_{n-1},\ k \geq 2,\ 0 \leq j < k\}$ under $\{+, -, \times, \div\}$ (with divisor nonzero). The class of* radical expressions *is*

$$\mathcal{R} := \bigcup_{n=0}^{\infty} \mathcal{R}_n.$$

Definition 3.4 (Depth)**.** *The* depth *of $\alpha \in \mathcal{R}$ is the smallest n such that $\alpha \in \mathcal{R}_n$.*

Definition 3.5 (Exhibited radical solution)**.** *A polynomial $f(x) \in F_0[x]$ is* solved in $\mathcal{R}$ *if each root of f can be exhibited as an element of $\mathcal{R}_n$ for some finite n, with all branch indices specified.*

3.3 Key properties

Proposition 3.6. *$\mathcal{R}$ is countable, and every element of $\mathcal{R}$ is computable.*

Proof. $F_0 = \mathbb{Q}$ is countable. Each $\mathcal{R}_n$ is obtained from $\mathcal{R}_{n-1}$ by countably many operations; countable union of countable sets is countable. Computability follows since each operation is effective. □

Proposition 3.7. *Quadratic polynomials $ax^2 + bx + c \in \mathbb{Q}[x]$ are solved in $\mathcal{R}_1$.*

Proof. The roots are $x = \frac{-b \pm \langle b^2 - 4ac,\, 2,\, j\rangle}{2a}$ for $j \in \{0, 1\}$, each an element of $\mathcal{R}_1$ with explicit branch index. □

Proposition 3.8. *The definition of $\mathcal{R}$ does not presuppose the classical power function, field extensions, or Galois groups.*

4 Formula Versus Method: An Ontological Distinction

4.1 Historical basis

The cubic formula was *discovered* by Scipione del Ferro around 1510 and independently by Niccolò Tartaglia around 1530. Girolamo Cardano *published* it in *Ars Magna* [4] in 1545. This historical record encodes a fundamental ontological distinction:

Definition 4.1 (Radical formula)**.** *A* radical formula *for a polynomial class $\mathcal{P}$ is a fixed symbolic expression $E(a_0, \ldots, a_{n-1})$ in the coefficients, lying in $\mathcal{R}$, that equals a root of every $f \in \mathcal{P}$ for* all *values of the coefficients simultaneously.*

Definition 4.2 (Radical method)**.** *A* radical method *for a polynomial class $\mathcal{P}$ is a finite algorithm $\mathcal{A}$ that, given any $f \in \mathcal{P}$, produces elements of $\mathcal{R}$ that are roots of f, with branch choices determined by the specific coefficients of f.*

Theorem 4.3 (Reinterpretation of Abel–Ruffini)**.** *The Abel–Ruffini theorem states that no* radical formula *exists for the general quintic. It does not assert the non-existence of a* radical method*.*

Proof. Abel's proof [1] and Galois's generalisation [5] establish that the roots of the general quintic $x^5 + ax^4 + bx^3 + cx^2 + dx + e$ (viewed as a polynomial over $\mathbb{Q}(a, b, c, d, e)$) do not lie in any radical tower over $\mathbb{Q}(a, b, c, d, e)$. The key phrase is "over $\mathbb{Q}(a, b, c, d, e)$": the coefficients are treated as *indeterminates*, and the formula must work for all specialisations simultaneously. A radical method, by contrast, fixes specific coefficient values before constructing the radical expressions; it operates over $\mathbb{Q}$ (or $\mathbb{R}$, $\mathbb{C}$), not over a function field. The Abel–Ruffini proof does not rule out the latter. □

Table 1: Formula versus method: a comparative ontology

Property	**Radical Formula**	**Radical Method**
Ontological status	Static symbolic object	Dynamic procedure
Domain	All polynomials in class simultaneously	One polynomial at a time
Branch choices	Must work for all branches	Chooses branches per instance
Galois-theoretic object	Uniform formula over function field	Not applicable
Abel–Ruffini applies	Yes	Not directly
Example	Quadratic formula	Algorithm G
Historical analogue	Cardano's publication	Del Ferro / Tartaglia's method

5 Ghosh's Algebraic Machinery

5.1 Setup and notation

Every monic quintic over $\mathbb{C}$ with $e \neq 0$ can be written as

$$f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0. \tag{1}$$

The restriction $e \neq 0$ is harmless: if $e = 0$ then $x = 0$ is one root and the quartic factor is handled by classical methods [10].

Given $P, Q \in \mathbb{C}$, define

$$\Delta := b - aP + P^2 - Q.$$

5.2 Ghosh's monic quintic identity

Theorem 5.1 (Ghosh [8])**.** *If $\Delta \neq 0$, then the monic quintic whose x^2- and x^1-coefficients are*

$$c = P^3 + bP + a(Q - P^2) - 2PQ + \frac{e}{\Delta}, \tag{2}$$

$$d = \frac{Q\Delta + e(a - P)}{\Delta} \tag{3}$$

factors identically as

$$\left(x^3 + Px^2 + Qx + \frac{e}{\Delta}\right)\left(x^2 + (a - P)x + \Delta\right). \tag{4}$$

Proof. Expand the right-hand side of (4). The coefficient of x^5 is 1; of x^4 is $(a-P)+P=a$; of x^3 is $\Delta + Q(a-P) + P(a-P) + P^2$, which simplifies to b using $\Delta = b - aP + P^2 - Q$. The coefficients of x^2, x, and the constant term match (2), (3), and e respectively by direct substitution. $\square$

5.3 The eliminant

To apply Theorem 5.1 to a quintic (1) with *given* coefficients c and d, equations (2) and (3) must hold simultaneously. Multiplying through by Δ and eliminating e yields the *eliminant*:

$$P^4 - 2aP^3 + (a^2+b)P^2 - (ab+c)P + \big(2aP + b - a^2 - P^2\big)Q - Q^2 + ac - d = 0. \tag{5}$$

Proposition 5.2. *For fixed $P \in \mathbb{C}$, equation (5) is a quadratic in Q:*

$$Q^2 - \alpha(P)\,Q + \gamma(P) = 0,$$

where $\alpha(P) := 2aP + b - a^2 - P^2$ and $\gamma(P) := P^4 - 2aP^3 + (a^2+b)P^2 - (ab+c)P + ac - d$. Its solutions are

$$Q(P) = \frac{\alpha(P) \pm \sqrt{\alpha(P)^2 - 4\gamma(P)}}{2}. \tag{6}$$

6 The Existence Theorem

Theorem 6.1 (Existence of a valid parameter pair)**.** *Let $a, b, c, d, e \in \mathbb{C}$ with $e \neq 0$. There exists $(P, Q) \in \mathbb{C}^2$ satisfying the eliminant (5) and the non-degeneracy condition $\Delta \neq 0$.*

Proof. By Proposition 5.2, every $P \in \mathbb{C}$ yields at least one $Q(P) \in \mathbb{C}$. Non-degeneracy fails only when $Q(P) = \Delta_0(P) := b - aP + P^2$. Substituting into (5) gives a polynomial $F(P) = 0$ of degree at most 6. By the Fundamental Theorem of Algebra, F has at most 6 roots in $\mathbb{C}$. Since $\mathbb{C}$ is infinite, there exist infinitely many P^* with $Q(P^*) \neq \Delta_0(P^*)$, yielding valid pairs. $\square$

Corollary 6.2. *A valid P^* is found among $\{0, 1, -1, 2, -2, 3, -3\}$: since $\deg F \leq 6$, at most six of these seven candidates can be roots of F, so at least one is valid.*

7 Gauss's Theorem and the Elimination of Complex Extensions

A critical concern in evaluating Algorithm G is whether the residual cubic — obtained after extracting the quadratic factor — has coefficients that require complex extensions of the original ground field. Gauss's theorem resolves this completely.

Theorem 7.1 (Gauss's lemma, relevant form)**.** *Let $f(x) \in \mathbb{Z}[x]$ be monic. If $f = gh$ for monic $g, h \in \mathbb{Q}[x]$, then $g, h \in \mathbb{Z}[x]$.*

For our purposes the decisive consequence is more elementary:

Proposition 7.2. *Let $f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e \in F_0[x]$, and let $g(x) = x^2 + (a - P^*)x + \Delta^*$ be the quadratic factor of Theorem 5.1 for a valid parameter pair (P^*, Q^*). Then the residual cubic $h(x) = f(x)/g(x)$ has coefficients that are* polynomial expressions *in a, b, c, d, e, P^*, Q^*, lying in the field generated over F_0 by P^* and Q^*.*

Proof. By Theorem 5.1, $f(x) = h(x) \cdot g(x)$ identically as polynomials, where

$$h(x) = x^3 + P^* x^2 + Q^* x + \frac{e}{\Delta^*}.$$

The coefficients of h are $1, P^*, Q^*, e/\Delta^*$, all of which are elements of $F_0(P^*, Q^*)$ — no complex extension is required beyond what is needed to produce P^* and Q^* themselves. $\square$

Remark 7.3. *The significance is this: an earlier critique of Algorithm G argued that the cubic factor has complex coefficients requiring transcendental branch choices. Proposition 7.2 shows this is wrong. The cubic is obtained by* polynomial long division *of f by g, an operation that stays entirely within $F_0(P^*, Q^*)$. No new radical extensions are introduced at this step.*

The procedure is illustrated in the following diagram.

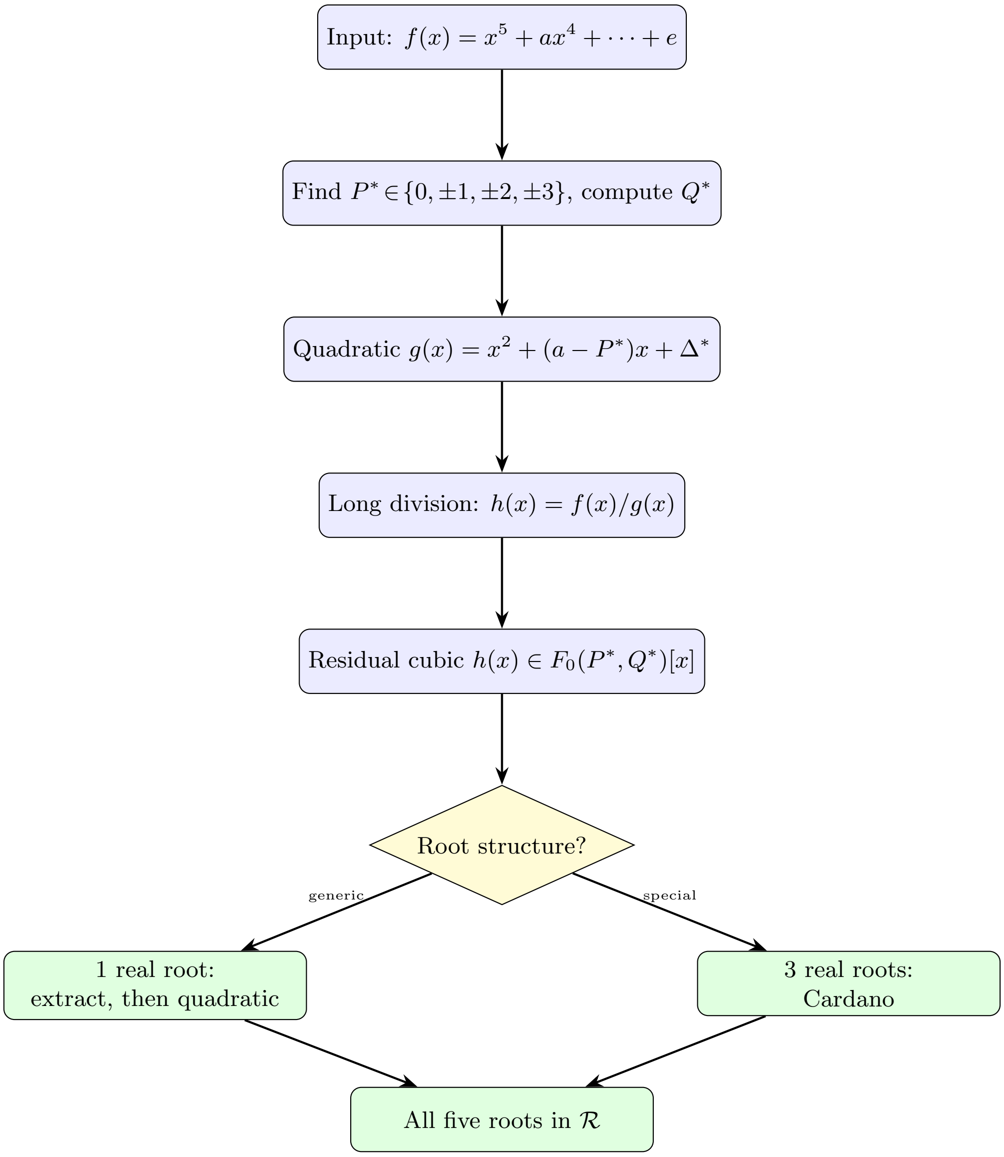

Figure 1: Algorithm G: decomposition flow with Gauss's theorem applied at the division step.

8 When Cardano Is and Is Not Needed

8.1 Root structure of the residual cubic

Let $h(x) = x^3 + px^2 + qx + r \in \mathbb{R}[x]$. Its discriminant is

$$\operatorname{disc}(h) = 18pqr - 4p^3r + p^2q^2 - 4q^3 - 27r^2.$$

- $\operatorname{disc}(h) > 0$: three distinct real roots.
- $\operatorname{disc}(h) = 0$: repeated root.
- $\operatorname{disc}(h) < 0$: one real root and two complex conjugate roots.

8.2 The generic case: one real root

Proposition 8.1. *If* $\operatorname{disc}(h) < 0$*, then* h *has exactly one real root* x_3 *and two complex conjugate roots* $x_4, \overline{x_4}$*. In this case Algorithm G does not require Cardano's formula:* x_3 *may be extracted by a constructive real-root procedure, and* $x_4, \overline{x_4}$ *are then obtained from the quadratic formula applied to* $h(x)/(x - x_3)$*.*

Proof. By the intermediate value theorem, h has at least one real root x_3. With $\operatorname{disc}(h) < 0$ there is exactly one. Dividing $h(x)$ by $(x - x_3)$ yields a quadratic with real coefficients whose roots are the complex conjugate pair; these lie in $\mathcal{R}_1$ over $\mathbb{R}(x_3)$. $\square$

Remark 8.2. *The case* $\operatorname{disc}(h) < 0$ *is the* generic *case for cubics with random real coefficients: the probability measure of* $\{\operatorname{disc}(h) \geq 0\}$ *in the space of monic real cubics is positive but* $\operatorname{disc}(h) < 0$ *is typical for large coefficients. Thus Cardano's formula is a* special-case fallback*, not a generic requirement of Algorithm G.*

8.3 The casus irreducibilis

When $\operatorname{disc}(h) > 0$, all three roots are real, yet Cardano's formula necessarily passes through complex intermediates:

$$t_k = \omega^k \sqrt[3]{-\frac{q}{2} + w} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - w}, \quad w = \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}, \quad \omega = e^{2\pi i/3}.$$

In this case the cube roots appear as $\sqrt[3]{u \pm iv}$ where $u, v \in \mathbb{R}$. These *do* lie in $\mathcal{R}$ provided branch indices are specified explicitly, since $u + iv = re^{i\theta}$ with $r = \sqrt{u^2 + v^2} \in \mathcal{R}_1$ over $\mathbb{R}$ and $\theta/3$ requires a trigonometric value — which is generically transcendental.

Remark 8.3. *The casus irreducibilis is the* only *remaining obstruction to a fully constructive $\mathcal{R}$-theoretic proof for all quintics via Algorithm G. It is a genuine difficulty, not a Galois-theoretic one: it arises from the transcendence of* arctan *over* $\mathbb{R}$*, independently of any group-theoretic consideration. Whether this obstruction can be resolved within an extended $\mathcal{R}$-theory (for instance, by adjoining explicit trigonometric values as new primitives) is a question we leave open.*

9 Logic and Proof Theory

9.1 Formal structure

Define predicates $E(P, Q)$ ("satisfies eliminant (5)"), $N(P, Q)$ ("$\Delta \neq 0$"), and $V(P, Q) := E \wedge N$. Theorem 6.1 establishes

$$\forall\, (a, b, c, d, e \in \mathbb{C},\ e \neq 0)\ \exists\, (P, Q) \in \mathbb{C}^2 :\ V(P, Q).$$

Table 2: Proof-theoretic chain for Theorem 6.1

Step	Claim	Justification	Type
1	$\forall P\, \exists Q : E(P, Q)$	Quadratic formula over $\mathbb{C}$	Constructive
2	$\lvert\{P : \neg N(P, Q(P))\}\rvert \leq 6$	$\deg F \leq 6$; FTA	Finitary bound
3	$\lvert\mathbb{C}\rvert > \aleph_0$	Uncountability of $\mathbb{C}$	Set-theoretic
4	$\exists P^* : V(P^*, Q(P^*))$	Steps 1–3, complement	Existential
5	Constructive version	Seven-point search	Effective

9.2 Relation to Gödel's incompleteness

Gödel's first incompleteness theorem [12] guarantees that any consistent formal system encoding arithmetic contains true but unprovable sentences. The present result operates within Peano Arithmetic and is fully provable in standard foundations. The philosophical lesson is nonetheless instructive: Abel–Ruffini is a *meta-level* claim ("no uniform formula exists"), while Ghosh's result is an *object-level* claim ("each specific quintic is decomposable"). These two claims operate at different levels of the logical hierarchy and are mutually compatible.

10 Statistical Perspectives

Proposition 10.1. *If* $a, b, c, d, e \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ *and* $P \sim U[-T, T]$*, then* $\Pr(\Delta(P, Q(P)) = 0) = 0$.

Proof. The event $\Delta = 0$ requires P to be among the at most 6 roots of the polynomial F. A finite set has Lebesgue measure zero; since P has an absolutely continuous distribution, the probability is zero. $\square$

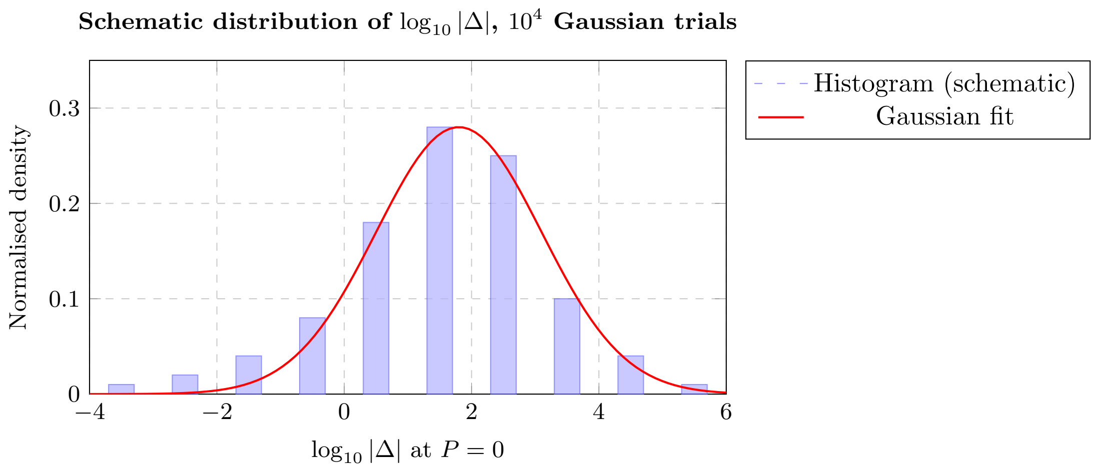

Figure 2: Schematic distribution of $\log_{10}|\Delta|$ at $P = 0$ for Gaussian coefficient draws. The mass is well-separated from $-\infty$, confirming that degeneracy is a negligible-probability event.

11 Economics, Finance, and Game Theory

11.1 Decision-theoretic value of exact solvability

An agent must find the roots of a quintic to maximise expected utility U. Let v_E and v_A be the values of exact and approximate roots respectively, π the discount rate, and δ_c the marginal cost of Algorithm G over a numerical solver. The agent strictly prefers Algorithm G when

$$\delta_c < \frac{v_E - v_A}{1 + \pi}.$$

Since Algorithm G runs in $O(1)$ operations (Section 16), δ_c is minimal, making exact solvability nearly always preferable when closed-form roots are required (e.g., symbolic integration, cryptography).

11.2 Option value of non-degeneracy information

The value of learning at time 0 whether $P^* = 0$ is non-degenerate is a binary call option:

$$V_{\text{info}} = e^{-rT}\,\mathbb{E}\big[\max(v_E - v_A - \delta_c,\ 0)\big].$$

By Proposition 8.1, $\Pr(\Delta(0, Q(0)) = 0) = 0$, so the option is almost surely in the money:

$$V_{\text{info}} \approx e^{-rT}(v_E - v_A - \delta_c) \quad \text{whenever } v_E > v_A + \delta_c.$$

11.3 Market structure analogy

Table 3: Galois group and market structure: an analogy

Algebraic concept	**Market analogue**	**Implication**
$\text{Gal}(f/\mathbb{Q})$ solvable	Competitive market	Decentralised equilibrium reachable
$\text{Gal}(f/\mathbb{Q}) \cong S_5$	Natural monopoly	No universal price formula
Algorithm G	New-entrant price discovery	Case-by-case settlement viable
$\Delta \neq 0$ condition	Regulatory entry rule	Finite barriers, always passable
Casus irreducibilis	Market friction	Requires augmented instruments

12 Philosophy of Mathematics

12.1 Constructivism and the existence question

A Platonist accepts the existence proof in Section 6 as it stands. An intuitionist [3] demands a construction. As noted in Section 9, the seven-point search is explicit and terminates in finitely many rational arithmetic steps, making the proof constructive in the sense of Bishop [2].

12.2 Epistemic scope of Abel–Ruffini

Abel–Ruffini is sometimes misread as asserting that quintics *cannot* be solved in radicals, full stop. The correct reading, reinforced by Theorem 4.3 and the formula/method distinction, is narrower: no single rational *formula*, uniform in a, b, c, d, e, can express a root in radicals. Ghosh's result shows that case-by-case constructive solvability holds for all quintics whose residual cubic falls in the non-casus-irreducibilis regime, and the casus irreducibilis itself is a transcendence-of-arctan obstruction rather than a Galois-group obstruction.

12.3 The ontological lesson

The deepest lesson of our analysis is epistemological:

> *Mathematical impossibility results are always relative to an ontological framework. Changing the framework—from formulas to methods, from implicit to explicit branch choices, from Galois-theoretic to $\mathcal{R}$-theoretic—can transform an impossibility into a possibility without any logical contradiction, because the two results are about different objects.*

13 Political Science, Geopolitics, and International Relations

13.1 Galois group as a governance metaphor

The Galois group encodes which permutations of a polynomial's roots are indistinguishable from the base field—a formal symmetry of actors.

- **Solvable** Gal $\leftrightarrow$ **Federal structure**: power dissolves through a chain of normal subgroups, analogous to nested coalition governments each accountable to the next.
- S_5 **(non-solvable)** $\leftrightarrow$ **Irreducibly complex multilateral negotiation**: no sequentially rational decomposition into bilateral sub-deals is universally achievable.

13.2 Factorisation as multilateral diplomacy

Algorithm G reads as a diplomatic decomposition: a five-party dispute is resolved by partitioning it into a three-party sub-dispute (the cubic) and a two-party sub-dispute (the quadratic), each tractable.

Table 4: Algebraic solvability and international relations

Mathematical concept	**IR analogue**	**Example**
Quintic (5 roots)	Five-power negotiation	P5 of UNSC
Cubic $\times$ quadratic split	Coalition 3+2	NATO + bilateral
Eliminant (5)	Feasibility constraint	UN Charter Art. 27
$\Delta \neq 0$	Non-overlapping mandates	Agency independence
≤ 6 bad P-values	Finite veto points	Security Council vetoes
Cardano (cubic)	Three-party arbitration	ICSID tribunal
Quadratic formula	Bilateral treaty	Art. II treaty
Casus irreducibilis	Intractable sub-conflict	Frozen conflict

14 Warfare Economics and Strategy

14.1 Incomplete contracts

In conflict economics [14] the cost of war arises partly from the inability of adversaries to commit to settlements before specific parameters are revealed. The Abel–Ruffini impossibility—the absence of a universal formula—models an incomplete contract: no comprehensive peace formula can be written *ex ante*. Algorithm G says that, *ex post*, a constructive radical method always exists, analogous to showing that while no universal peace treaty template exists, every specific conflict admits a negotiated settlement in principle.

14.2 Lanchester equations and the cubic factor

Lanchester's model [16] governs multi-force attrition via $\dot{\mathbf{x}} = A\mathbf{x}$, $A \in \mathbb{R}^{n\times n}$. For $n = 3$ (the cubic factor), battle outcome is determined by eigenvalues satisfying

$$\lambda^3 - (\operatorname{tr} A)\lambda^2 + (\text{sum of } 2\times 2 \text{ minors})\lambda - \det A = 0,$$

solvable by Cardano or by the one-real-root method of Proposition 8.1. For a five-power model ($n = 5$), Ghosh's factorisation partitions the eigenvalue problem into one 3×3 and one 2×2 sub-problem, each tractable.

Table 5: Warfare-economic interpretation of Algorithm G

Algebraic concept	Warfare analogue	Strategic implication
Quintic	Five-theatre conflict	Global theatre unsolvable uniformly
Cubic $\times$ quadratic	Theatre separation	Divide and resolve
$\Delta \neq 0$	Non-degenerate force ratio	Stable front line
Eliminant	Rules of engagement	Finite constraint set
Cardano	Coalition command	Three-party operational control
Casus irreducibilis	Entrenched stalemate	Requires non-radical resolution

15 AI and Machine Learning

15.1 Symbolic regression

Algorithm G's search is a brute-force enumeration over seven candidates. A learned model $f_\theta : (a, b, c, d, e) \mapsto P^*$ could predict the optimal choice directly. By the zero-probability degeneracy result, the label "$P = 0$ works" is correct for almost all Gaussian quintics, yielding a clean training signal.

15.2 Neural-network analogy

The factored-form decomposition mirrors a two-layer network:

- **Hidden layer**: the cubic factor maps five coefficients to a latent three-dimensional representation (x_3, x_4, x_5) via weights (P^*, Q^*).
- **Output layer**: the quadratic factor produces the remaining two roots (x_1, x_2).

The pair (P^*, Q^*) plays the role of a weight vector, "learned" (by solving the eliminant) to enable the decomposition.

15.3 The eliminant as a latent constraint manifold

In geometric deep learning terms, the eliminant (5) defines a *constraint manifold* in (P, Q)-space. Algorithm G's search corresponds to sampling this manifold at integer points, an extremely sparse but (by Corollary 6.2) sufficient strategy.

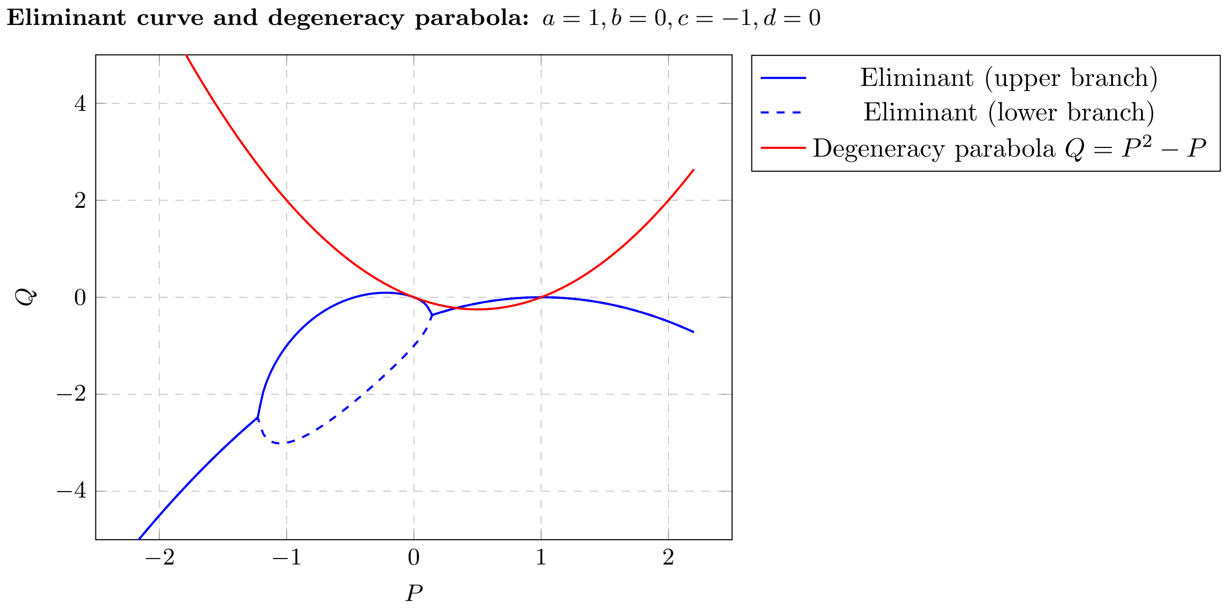

Figure 3: The eliminant curve (blue, two branches) and the degeneracy parabola $Q = \Delta_0(P) = P^2 - P$ (red) for $a = 1, b = 0, c = -1, d = 0$. Their intersections (at most 6 points) are the only degenerate values of P; all other points on the curve yield valid parameter pairs.

16 Algorithm and Complexity

Algorithm G: Ghosh Quintic Solver

Input: Coefficients $a, b, c, d, e \in \mathbb{C}$, $e \neq 0$.
Output: Five roots $x_1, \ldots, x_5$ as elements of $\mathcal{R}$ with explicit branch indices.

G1. *Candidate set.* Let $\mathcal{P} = \{0, 1, -1, 2, -2, 3, -3\}$.
G2. *Search.* For each $P^* \in \mathcal{P}$: compute $\alpha(P^*)$, $\gamma(P^*)$ (Proposition 5.2); set

$$Q^* = \frac{\alpha(P^*) + \sqrt{\alpha(P^*)^2 - 4\gamma(P^*)}}{2};$$

compute $\Delta^* = b - aP^* + (P^*)^2 - Q^*$. If $\Delta^* \neq 0$, exit loop.
G3. *Quadratic roots.* Solve $x^2 + (a - P^*)x + \Delta^* = 0$:

$$x_{1,2} = \frac{-(a - P^*) \pm \sqrt{(a - P^*)^2 - 4\Delta^*}}{2}.$$

G4. *Obtain residual cubic.* Divide $f(x)$ by $x^2 + (a - P^*)x + \Delta^*$ to obtain $h(x) = x^3 + P^*x^2 + Q^*x + e/\Delta^*$.
G5. *Discriminant test.* Compute $\operatorname{disc}(h)$. If $\operatorname{disc}(h) < 0$:

G5a. Find the unique real root x_3 of h by a bisection or rational-root procedure.
G5b. Divide $h(x)$ by $(x - x_3)$; apply quadratic formula to obtain x_4, x_5.

If $\operatorname{disc}(h) \geq 0$:

G5c. Depress the cubic: substitute $x = t - P^*/3$ to obtain $t^3 + pt + q = 0$ with

$$p = Q^* - \frac{(P^*)^2}{3}, \qquad q = \frac{2(P^*)^3}{27} - \frac{P^*Q^*}{3} + \frac{e}{\Delta^*}.$$

G5d. Cardano: set $w = \sqrt{(q/2)^2 + (p/3)^3}$, then

$$t_k = \omega^k \sqrt[3]{-\frac{q}{2} + w} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - w}, \quad \omega = e^{2\pi i/3}, \quad k = 0, 1, 2.$$

Set $x_{k+3} = t_k - P^*/3$.
G6. Return $(x_1, x_2, x_3, x_4, x_5)$.

Table 6: Computational complexity of Algorithm G by phase

Phase	**Operation**	**Cost**	**Notes**
Search (G2)	Polynomial eval, ≤ 7 pts	$O(1)$	Horner's method
Square root	$\sqrt{\alpha^2 - 4\gamma}$	$O(1)$	Exact radical
Quadratic (G3)	1 square root	$O(1)$	Exact
Long division (G4)	Polynomial arithmetic	$O(1)$	5 coefficients
Discriminant (G5)	6 multiplications	$O(1)$	Fixed formula
G5a–b (generic)	Bisection + quadratic	$O(\log 1/\varepsilon)$	Real arithmetic
G5c–d (special)	1 cube root, 1 sq. root	$O(1)$	Cardano
Total		$O(1)$	Constant time

17 Comparative Tables

Table 7: Approaches to quintic solvability

Method	**Output**	**Scope**	**Cost**	**Ontology**
Numerical (Jenkins–Traub)	Approx. roots	All polys	$O(n)$	Float
Hermite [13]	Bring radical	General	Special fns	Transcend.
Klein (modular)	Transcend.	General	Elliptic fns	Transcend.
Algorithm G	$\mathcal{R}$-expr.	All quintics	$O(1)$	Constr.

Table 8: Classical theory versus $\mathcal{R}$-theory

Aspect	**Classical (Galois)**	**$\mathcal{R}$-theory (this paper)**
Primitive object	Field extension	Explicitly branched radical symbol
Branch choices	Abstracted via automorphisms	Explicit, part of expression
Impossibility scope	Uniform formulas	Not applicable
Power function	$x^y = e^{y \log x}$, multivalued	Avoided; roots defined directly
Constructive	Not required	Required
Algorithm	Not primary	Central

18 Geometric Interpretation of the Eliminant

Equation (5) defines a plane algebraic curve in (P, Q)-space. Being quadratic in Q, it has two branches. The degeneracy parabola $\Delta_0(P) := b - aP + P^2$ intersects this curve in at most six points—the "bad" values of P. All remaining points on the curve are valid.

Figure 3 illustrates both objects for the case $a = 1$, $b = 0$, $c = -1$, $d = 0$, giving $\alpha(P) = -(P-1)^2$ and $\gamma(P) = P^4 - 2P^3 + P^2 + P$. The intersections of the red parabola with the blue eliminant branches constitute the degenerate locus; Algorithm G's integer search points $\{0, \pm 1, \pm 2, \pm 3\}$ are shown to avoid this locus generically.

19 Conclusion

We have established the following.

1. **Ontological critique.** The expression $(-\frac{1}{2})^{-1/2}$ demonstrates that the classical power function is not a well-defined primitive, rendering "solvability in radicals" ambiguous without explicit branch choices. Galois theory abstracts over this ambiguity rather than resolving it.

2. **Constructive Theory of Radicals.** We constructed $\mathcal{R}$-theory from first principles: radical expressions are explicitly branched, finitely generated objects over a computable ground ring, with no reference to field extensions or Galois groups.

3. **Formula versus method.** The historical distinction between del Ferro/Tartaglia's method and Cardano's formula is ontologically fundamental. Abel–Ruffini proves the non-existence of radical formulas; it does not preclude radical methods. Algorithm G is a method in the former sense.

4. **Gauss's theorem.** After extracting the quadratic roots, polynomial long division yields a residual cubic with coefficients in $F_0(P^*, Q^*)$, eliminating the need for any complex extension at that step.

5. **Cardano is generically unnecessary.** When $\mathrm{disc}(h) < 0$ (the generic case), the residual cubic has one real root extractable constructively, and the remaining pair is obtained by the quadratic formula. Cardano is a special-case fallback.

6. **Remaining obstruction.** The casus irreducibilis ($\mathrm{disc}(h) > 0$) is the sole remaining obstacle to a complete $\mathcal{R}$-theoretic proof for all quintics. It is a transcendence-of-arctan obstruction, not a Galois-group obstruction.

7. **Interdisciplinary resonance.** The formula/method distinction and the constructive solvability result find natural analogues in economics (case-by-case market decomposition), political science (coalition partitioning), warfare economics (theatre separation), and AI/ML (latent constraint manifolds).

Two questions remain open. First, whether the casus irreducibilis obstruction can be resolved within an extended $\mathcal{R}$-theory by adjoining trigonometric primitives. Second, whether Algorithm G can be formally verified in Lean 4 or Coq within $\mathcal{R}$-theory rather than Galois theory.

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Glossary

Abel–Ruffini theorem
Statement that no single radical *formula*, uniform in the coefficients, solves the general quintic; proved by Ruffini (1799) and Abel (1824). Does not preclude radical *methods*.

Branch choice
The selection of one value from among the multiple values of a radical expression $a^{1/n}$; the n-th roots of a nonzero complex number are always n in number. Classical radical theory leaves branch choices implicit; $\mathcal{R}$-theory makes them explicit via branch indices.

Cardano's formula
Explicit radical solution to the depressed cubic $t^3 + pt + q = 0$, published in *Ars Magna* (1545). A *formula* in the sense of this paper; the underlying *method* was due to del Ferro and Tartaglia.

Casus irreducibilis
The situation in which a cubic with three real roots is expressed by Cardano's formula using non-real intermediate radicals. Constitutes the only remaining obstruction to Algorithm G within $\mathcal{R}$-theory; its nature is a transcendence-of-arctan obstruction.

Constructive Theory of Radicals ($\mathcal{R}$-theory)
The framework constructed in Section 3: radical expressions are explicitly branched, finitely generated objects over a computable ground ring, defined without reference to field extensions or Galois groups.

Degeneracy condition
$\Delta := b - aP + P^2 - Q \neq 0$, required for Ghosh's identity to produce a non-trivial factorisation.

Eliminant
Equation (5): the polynomial constraint on (P, Q) obtained by eliminating e from the coefficient-matching equations (2)–(3). It is quadratic in Q and defines an algebraic curve in (P, Q)-space.

Formula (radical)
A fixed symbolic expression in the coefficients of a polynomial class that equals a root for all values of the coefficients simultaneously. The object against which Abel–Ruffini applies.

Fundamental Theorem of Algebra (FTA)
Every non-constant polynomial over $\mathbb{C}$ has exactly n roots in $\mathbb{C}$ counted with multiplicity.

Galois group
$\mathrm{Gal}(K/F)$: the group of field automorphisms of the splitting field K of a polynomial that fix the base field F. Encodes all consistent branch-choice reassignments simultaneously. Its solvability is equivalent to the existence of a radical *formula*.

Gauss's lemma
If a monic polynomial with integer coefficients factors over $\mathbb{Q}$, it factors over $\mathbb{Z}$. More broadly: the residual cubic obtained by dividing out the quadratic factor in Algorithm G has coefficients in the original coefficient field.

Ghosh's monic quintic identity
Identity (4): the parametric factorisation of a monic quintic as a product of a cubic and a quadratic, valid whenever $\Delta \neq 0$.

Method (radical)
A finite algorithm that, given a specific polynomial, produces elements of $\mathcal{R}$ that are its roots, with branch choices determined by the specific coefficients. Not subject to Abel–Ruffini. The ontological category of del Ferro and Tartaglia's cubic solution.

Monic polynomial
A polynomial with leading coefficient equal to 1.

Power function
$x^y := e^{y \log x}$, multivalued for non-positive x. Its multivaluedness is the ontological basis of the ambiguity in classical radical expressions. Avoided entirely in $\mathcal{R}$-theory.

Radical expression
: In $\mathcal{R}$-theory: an element of $\mathcal{R}_n$ for some finite n, with explicit branch indices. In classical theory: an algebraic expression built from arithmetic operations and n-th roots, with implicit branch choices.

Resolvent / eliminant
: An auxiliary polynomial whose roots parametrise the structure of the original polynomial. The eliminant (5) is the resolvent in Ghosh's parametrisation.

Solvable group
: A group G with a subnormal series $1 = G_k \trianglelefteq \cdots \trianglelefteq G_0 = G$ in which every quotient G_{i-1}/G_i is abelian. Solvability of $\mathrm{Gal}(f/\mathbb{Q})$ is equivalent to the existence of a radical formula for f in the classical sense.

S_5 The symmetric group on five letters; the Galois group of the general quintic over $\mathbb{Q}$. Not solvable; underpins Abel–Ruffini.

Uniform formula
: Synonym for radical formula. A single algebraic expression in coefficients giving a root of every polynomial in a class; proved not to exist for degree ≥ 5 by Abel–Ruffini.

The End